Ting-Yu Tsai (no sponsorship required, permanent resident)

Summary

I am a Ph.D. student in Computer Science at the University at Albany specializing in computer vision, machine learning, multimodal AI, and medical AI. Experienced in conducting advanced research and developing deep learning systems for medical imaging, video understanding, and intelligent healthcare applications. Brings strong technical expertise, analytical problem-solving skills, and a proven ability to translate cutting-edge research into reliable and scalable real-world solutions.

Education

- **University at Albany - SUNY** — *NY, USA*
  Ph.D. in Computer Science; GPA: 3.73/4.0 — *Aug. 2023 - present*
- **National Taiwan University** — *Taipei, Taiwan (R.O.C.)*
  Master in Biomedical Electronics and Bioinformatics; GPA: 4.11/4.3 — *Aug. 2019 - Jul. 2021*
- **Chung Shan Medical University** — *Taichung, Taiwan (R.O.C.)*
  Bachelor in Department of Medical Informatics; GPA: 4.21/4.3 — *Aug. 2015 - Jul. 2019*

Experience

- **Research Assistant** — University at Albany - SUNY — *NY, USA* — *Jan. 2025 - May. 2025*
  - **Mamba-2 Quantization and Analysis:** Researched Mamba-2 quantization and activation behavior by adapting Quamba2 for fake quantization, profiling inference efficiency, and analyzing SSM distributions and eigenvalue patterns to investigate model stability and output outliers.
- **Research Assistant** — University at Albany - SUNY — *NY, USA* — *Jan. 2024 - Dec. 2024*
  - **Cardiovascular Image Segmentation:** Developed and benchmarked deep learning models for cardiovascular image segmentation using Python and PyTorch, integrating Sharpness-Aware Minimization (SAM) and uncertainty-aware loss design to improve generalization, robustness, and performance across challenging medical imaging tasks.
- **Process Engineer** — Taiwan Semiconductor Manufacturing Co., Ltd. (TSMC) — *Tainan, Taiwan (R.O.C)* — *Nov. 2021 - Jun. 2022*
  - Tested and monitored manufacturing equipment to support stable semiconductor production, improved system processes to enhance operational efficiency and reliability, conducted risk assessments to maintain quality and safety standards, and collaborated with cross-functional teams to resolve process and equipment-related challenges.
- **Software Engineer Intern** — Test Research Inc. — *Taipei, Taiwan (R.O.C)* — *Aug. 2019 - Jul. 2021*
  - **Super Resolution for X-Ray Images:** Developed and optimized a deep learning model for X-ray image super-resolution using Python, TensorFlow, and TensorLayer, improving image quality and enlargement efficiency and outperforming five classic AI models and four interpolation methods in both quality and computation speed.
- **Software Engineer Intern** — Chung Shan Medical University Hospital — *Taichung, Taiwan (R.O.C)* — *Jul. 2018 - Aug. 2018*
  - **Automatic Head and Neck CT Image Organ Contour Selection System:** Developed a Python-based automated organ contouring system for head and neck CT images, building an end-to-end workflow for CT image conversion, organ segmentation, and RS file generation, applying GAN-based deep learning to improve workflow consistency, and reducing organ selection and contouring time by 70% for clinical staff.

Projects

- **Rat Seizure Detection & Classification** [1]
  Developed an AI-powered framework for automated rat seizure detection and stage classification from video, combining deep learning and computer vision to support quantitative behavioral analysis in epilepsy research. The system reduces reliance on slow and subjective manual observation and demonstrates a scalable approach for preclinical seizure monitoring and neurological behavior studies. (2026)
- **ReDiPrune: Relevance-Diversity Pre-Projection Token Pruning for Efficient Multimodal LLMs** [2]
  Developed ReDiPrune, a training-free visual token pruning method for improving multimodal LLM efficiency by pruning tokens before the vision-language projector. Designed a lightweight relevance-diversity scoring strategy to preserve query-relevant and non-redundant visual information, enabling plug-and-play integration into existing MLLM pipelines without retraining or architectural changes. (2026)
- **IntelliCardiac – Cardiac MRI Segmentation and Disease Classification Platform** [3]
  Developed an end-to-end full-stack medical imaging platform for automated cardiac MRI segmentation and five-class disease classification with real-time visualization and workflow-ready deployment. The segmentation module achieved 92.6% overall accuracy, while the two-stage classification pipeline reached 98% accuracy, outperforming prior integrated systems. Tech: Python, PyTorch, JavaScript, Node.js, MongoDB, Medical Imaging, Deep Learning. (2025)
- **An Uncertainty-Aware Generalization Framework for Cardiovascular Image Segmentation** [4]
  Developed and benchmarked UU-Mamba, a reliable cardiovascular image segmentation framework that integrates SAM and an uncertainty-aware composite loss to improve robustness and generalization, achieving state-of-the-art performance on ACDC, ImageCAS, and Aorta datasets. (2024)
- **Automatic Image Exposure Classification and Enhancement System**
  Built a deep learning-based system for automatic image exposure classification and enhancement, achieving 0.91 classification accuracy and improving image quality to a level comparable to camera-captured results. (2021)

Publications

[1] **Ting Yu Tsai**, A. Yu, et al., "RatSeizure: A Benchmark and Saliency-Context Transformer for Rat Seizure Localization," *arXiv preprint arXiv:2603.26780*, 2026.

[2] A. Yu, **Ting Yu Tsai**, Z. Zhang, W. Lu, F. X. .-. Ye, and M.-C. Chang, "ReDiPrune: Relevance-Diversity Pre-Projection Token Pruning for Efficient Multimodal LLMs," *arXiv preprint arXiv:2603.24680*, 2026.

[3] **Ting Yu Tsai**, A. Yu, et al., "IntelliCardiac: An Intelligent Platform for Cardiac Image Segmentation and Classification," in *2025 IEEE 8th International Conference on Multimedia Information Processing and Retrieval (MIPR)*, 2025, pp. 109–115.

[4] **Ting Yu Tsai**, L. Lin, S. Hu, M.-C. Chang, H. Zhu, and X. Wang, "UU-Mamba: Uncertainty-aware U-Mamba for Cardiac Image Segmentation," in *2024 IEEE 7th International Conference on Multimedia Information Processing and Retrieval (MIPR)*, 2024, pp. 267–273.

[5] **Ting Yu Tsai**, L. Gui, et al., "An Uncertainty-Aware Generalization Framework for Cardiovascular Image Segmentation," *arXiv preprint arXiv:2409.14305*, 2026.

[6] X. Wang, **Ting Yu Tsai**, et al., "Spotting the Fakes: A Deep Dive into GAN-Generated Face Detection," *ACM Transactions on Multimedia Computing, Communications and Applications*, vol. 21, no. 7, pp. 1–24, Jul. 2025.

[7] **Ting-Yu Tsai**, Y.-J. Hsieh, H.-C. Tseng, C.-L. Chin, and Y.-H. Shao, "Based on DICOM RT Structure and Multiple Loss Function Deep Learning Algorithm in Organ Segmentation of Head and Neck Image," in *Future Trends in Biomedical and Health Informatics and Cybersecurity in Medical Devices*, 2020, pp. 428–435.

[8] **Ting-Yu Tsai**, C.-L. Chin, M.-C. Chin, and W.-E. Chen, "Facial skin image classification system using Convolutional Neural Networks deep learning algorithm," in *2018 9th International Conference on Awareness Science and Technology (iCAST)*, 2018, pp. 51–55.

Honors and Awards

- **Third Prize**, AI and Robotics Creative Design Competition — 2018
- **Honorable Mention**, Creative Dredging Engineering & Monitoring Technology Competition — 2018
- **Honorable Mention**, National Creativity Competition of Electronic Design (14th) — 2018
- **Honorable Mention**, Creative Software Application Competition — 2018
- **First Prize and Best Popularity Award**, Taipei Medical University Venture Capital Investment Contest — 2017
- **First Prize**, International Conference on Medical and Health Informatics, for "Incidence and Clinical Features of Multiple Primary Malignant Neoplasms: A 5-Year Retrospective Analysis" — 2017
- **Gold Prize**, Korea International Women's Invention Exposition (KIWIE) — 2013
- **German Special Prize**, Korea International Women's Invention Exposition (KIWIE) — 2011

Skills

- **Research Areas:** Computer Vision, Machine Learning, Multimodal AI, Medical AI, Medical Imaging Analysis
- **Programming Languages:** Python, C/C++, MATLAB, PHP, JavaScript, HTML, CSS, VBA
- **Machine Learning & AI:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas
- **Libraries & Tools:** Dicompyler, TCPDF, PHPExcel, OpenSSL
- **Databases:** MySQL